MILITARY SPECIFICATION

CART, TEST EQUIPMENT

This specification has been approved by the Naval Air Systems Command, Department of the Navy.

## 1. SCOPE

1.1 Scope - The equipment covered by this specification shall be for Organizational Level usage and shall be a lightweight, mobile, special support equipment cart for the mounting and operation of the RF Noise Analyzer Electronic Units.

1.2 Classification - The Test Equipment Cart shall consist of the following items and shall hereinafter be referred to as the Cart.

| Item | Applicable Paragraph |
|---|---|
| Chassis | 3.4.1 |
| Running Gear | 3.4.2 |
| Cable Harness | 3.4.3 |
| Cable Assemblies | 3.4.4 |
| Waveguide Assemblies | 3.4.5 |

1.3 Associated Equipment - The Cart shall operate with the following associated equipment which shall not be supplied as part of the Cart:

(a) Electronic Units, RF Noise Analyzer, (411R702G01) (hereinafter referred to as Electronic Units)

(b) Cooler, Air-Hydraulic Unit, HD-747/AWM-22

(c) Cart, Maintenance, V-387/AWM-22





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## 2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

| | |
|---|---|
| MIL-C-17 | Cables, Radio Frequency; Coaxial, Dual Coaxial, Twin Conductor, and Twin Lead |
| MIL-C-71 | Connectors, "N" for Radio Frequency Cables |
| MIL-W-85 | Waveguides, Rigid, Rectangular, General Specification for |
| MIL-F-3922 | Flanges, Waveguide, General Purpose, General Specification for |
| MIL-E-5422 | Testing, Environmental, Aircraft, General Specification for |
| MIL-I-6181 | Interference Control Requirements, Aircraft Equipment |
| MIL-M-8090 | Mobility, Towed Aerospace Ground Equipment, General Requirements for |
| MIL-H-15362 | Harnesses; Test Bench, Avionic Equipment, General Specification for |
| MIL-W-16878 | Wire, Electrical, Insulated, High Temperature |
| MIL-E-17555 | Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of |
| MIL-T-18303 | Test Procedures; Preproduction and Acceptance for Aircraft Electronic Equipment, Format for |
| MIL-T-21200 | Test Equipment for use with Electronic and Fire Control Systems, General Specification for |
| MIL-T-23336 | Technical Directives (Letter Type): Preparation of |



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STANDARDS

Military

| | |
|---|---|
| MIL-STD-108 | Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment |

DRAWINGS

Military

| | |
|---|---|
| MS 3120 | Connectors, Receptacle, Electric, Crimp-Type |
| MS 3126 | Connector, Plug, Electrical |
| MS 24663 | Connector, Plug, Electrical - With Spring Loaded Pivoted Grounding Blade, Polarized |
| MS 35168 | Connector, Plug UG-88E/U |

Westinghouse Electric Corporation (97942)

| | |
|---|---|
| 411R702 | Key Drawing, Electronic Units, RF Noise Analyzer |
| 575R516 | Connector, Plug, Pigmy E Type |
| 577R569 | Connector, Electrical, Coaxial |
| 577R590 | Connector, Receptacle, Electrical |
| 577R831 | Connector, Receptacle, Polarized, Duplex |
| 579R831 | Connector, Electrical, Coaxial Plug, Type BSM |
| 580R473 | Connector, Electrical, Coaxial |

2.1.1 Availability of Documents - Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting office. See 6.4.

3. REQUIREMENTS

3.1 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing the following shall govern:



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(a) Parts and Materials as approved by MIL-T-21200 shall be given first consideration.

(b) Guidelines for the selection of nonstandard parts shall be in accordance with MIL-T-21200.

(c) In the selection of parts, the above considerations shall apply if the assembly is to be built for detail part maintenance. In the event the entire assembly is built to be expended, the assembly shall be constructed and the quality of the individual parts shall be such that the unit as a whole will pass electrical and environmental test.

3.1.1 Nonstandard Parts - Nonstandard parts and materials shall be as outlined in MIL-T-21200, Materials, Parts and Processes, except that mechanical parts shall be considered as nonstandard parts only when the material requirements of MIL-T-21200, as modified herein, are not met.

3.1.2 Noncorrosive Hardware - External Military Standard (MS) hardware which is used during servicing, accessing or installing test equipments shall be of a corrosive-resistant type and shall not be further finish-treated.

3.2 Design and Construction - The Cart, excluding the Cable Harness (3.4.3), shall conform with all the applicable requirements of MIL-T-21200 and the Cable Harness shall conform with all the applicable requirements of MIL-H-15362 for design, construction and workmanship, unless otherwise specified herein.

3.2.1 Total Weight - The total empty weight of the Cart, including all cabling shall not exceed 700 pounds. The gross weight with all Electronic Units installed shall not exceed 1020 pounds.

3.2.2 Cables and Connectors - The Cart shall provide for the use of cables and connectors in accordance with MIL-T-21200 and MIL-H-15362, as applicable.

3.2.3 Interchangeability - The Cart shall meet the interchangeability requirements as defined in MIL-T-21200.

3.2.4 Interference Control - With all Electronic Units installed, the generation of radio interference by the Cart and the vulnerability of the Cart to radio interference shall be within the limits of MIL-I-6181.



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3.2.5 Maintenance Provisions and Field Testing - Provisions for maintenance shall be as specified in MIL-T-21200.

3.2.6 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory tests.

| | |
|---|---|
| Temperature | Room Ambient: 30° ±10° Centigrade (C)<br>86° ±18° Fahrenheit (F) |
| Altitude | Normal Ground |
| Vibration | None |
| Humidity | Room ambient up to 90 percent relative humidity |

3.2.7 Service Conditions - The Cart shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in MIL-T-21200 for Class 2 equipment. The Cable Harness shall meet the environmental service conditions of MIL-H-15362.

3.2.7.1 Weatherproof - The Cart, with covers and doors closed, shall be splashproof per MIL-STD-108.

3.3 Performance

3.3.1 General - Unless otherwise specified, values set forth to establish the requirements for satisfactory performance apply to performance under both standard and extreme service conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified.

3.3.2 Function - The Test Equipment Cart shall provide the mounting, interconnection, and means of transportation for the Electronic Units at the Organizational Level of maintenance. When connected to a source of prime power, it shall permit operation of the Electronic Units for detailed maintenance of the AN/AWG-10, AN/AWG-11 and AN/AWG-12 Missile Control Systems (MCS).

3.3.3 Form Factor - The Cart shall be a four-wheel, towable vehicle of rectangular shape with a bottom truncated center section across the width of the unit. The center section shall have a minimum of 5 inches ground clearance at maximum loaded weight. The Cart main frame shall not



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exceed 78 inches by 36 inches by 36 inches. With tow bar extended in the towing position, the length shall not exceed 126 inches.

3.3.4 Mobility - The Cart shall be of the type classified as Type I, Class 1, Group C, towed support equipment, per MIL-M-8090.

3.3.5 Lighting - Lighting shall be provided to illuminate the front panels of those Electronic Units installed in the right forward section of the Chassis (3.4.1).

3.3.6 Loading - The Cart shall be capable of supporting a loading of 250 pounds over a 10 square inch area without suffering permanent deformation.

3.3.7 Acoustic Environment - The Cart shall provide protection for satisfactory operation of the contained Electronic Units in an acoustic noise environment of 110 decibels (db) (referenced to 0.0002 microbar level).

3.4 Detail Requirements

3.4.1 Chassis

3.4.1.1 Function - The Chassis shall provide the enclosure and mounting structure for the Electronic Units and shall furnish protection to the Electronic Units necessary to meet the Service Conditions (3.2.7).

3.4.1.2 Form Factor - The Chassis shall be rectangular in shape with a truncated bottom center section extending the width of the unit. Dimensions shall not exceed 78 inches by 36 inches by 31 inches. The right forward section shall have a sloping, recessed framework to accept two rows of standard 19 inch type drawers. This section shall be enclosed by a hinged, accordian-fold type top cover. The right aft section shall have an enclosure to mount the RF Panel of the Electronic Units on slide rails. The enclosure shall have an acoustically shielded interior. An access door shall cover the opening to the RF Panel enclosure. The bottom center section shall provide the mounting enclosure for the Power Supply Drawer of the Electronic Units. Access to the Power Supply Drawer enclosure shall be from the left side of the chassis. Provisions shall be made for a drain in the bottom of the Chassis to permit removal of moisture. Handles shall be provided on the front and rear of the Chassis to facilitate lateral Cart movement.

3.4.1.3 Cover - A Cover shall be provided to afford weatherproof and physical protection to equipments mounted in the right forward recessed section. The Cover shall be of the folding accordian type and shall hinge at the midsection of the top surface of the Chassis. It shall extend from the



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hinge line down the right side of the Chassis to the bottom of the recessed section. The Cover shall be secured in the closed position by quick disconnect fasteners at the bottom. Tiedown fasteners shall be provided for the Cover when in the open, folded position on top of the Chassis.

3.4.1.4 Access Door - The Access Door shall attach to the left aft side of the Chassis and shall close the opening of the RF Panel enclosure. It shall have acoustic shielding material on its inner face.

3.4.1.5 Storage - The Chassis shall provide stowage space for cables, technical manuals, operating and servicing manuals and miscellaneous accessories. Three, pull down stowage compartments shall be provided on the left side of the Chassis and shall have restraining fastenings. Fastenings shall be of a releasable type to permit the stowage compartments to be fully lowered. A compartment shall also be provided on the front of the Chassis with a sloping top cover and a hasp for locking purposes.

3.4.1.6 Tiedown Provisions - The Chassis shall have two tiedown rings on the front which shall be compatible with standard aircraft tiedown devices and two tiedown attachment points on the rear Bumper (3.4.1.9).

3.4.1.7 Lighting - Lighting shall be provided through a translucent and a red filter lens and shall illuminate the Electronic Units panels when installed. Lighting shall also be provided for use with servicing instructions and technical manuals at the Cart under nighttime operating conditions.

3.4.1.8 Reflectors - Reflectors shall be located on both sides and on the front and rear of the Chassis. Two red reflectors shall be mounted on the rear. Two amber reflectors shall be mounted on each side and on the front.

3.4.1.9 Bumper - A Bumper shall be attached to the rear of the Chassis at the bottom to protect the Running Gear (3.4.2), the Chassis and installed units from rear end collision. One tiedown attachment point shall be provided at each end of the Bumper.

3.4.1.10 Electrical Connections

3.4.1.10.1 Power Connector - A receptacle, MS 3120E12-3P, shall be provided on the rear of the Chassis for single-phase, 400 cycles per second (cps) primary power to the Cart.

3.4.1.10.2 Ground Connection - A grounding stud shall be provided.



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3.4.1.11 Cable Mounting Fixture - A drum type, nonrotating fixture shall be provided on the rear of the Chassis for cable stowage.

3.4.2 Running Gear

3.4.2.1 Function - The Running Gear shall provide the required Mobility (3.3.4) to the Cart.

3.4.2.2 Form Factor - The Running Gear shall consist of four caster assemblies, tie rods and linkages and semipneumatic tires. The two rear caster assemblies shall be locked directly to the main Chassis while the two forward caster assemblies shall be linked to the towbar. All caster assemblies shall be capable of swiveling a minimum of 360 degrees by the release of manually operated locking pins. Tires shall be semipneumatic, shall be 8-3/4 inches rim size and shall have a nominal 3 inch tread width. Brakes shall be provided on each of the four wheels of the caster assemblies and shall be of the internal, expanding shoe, parking type. Vertical movement of the towbar shall mechanically control the braking action. Deadman braking shall lock all four wheels when the towbar is dropped to the deck or raised to the vertical stow position. Brakes shall be released when the towbar is in the horizontal, towing position. Two tie rods and a steering linkage shall connect the two forward caster assemblies and the towbar for steering purposes. A 1:1 turning ratio shall exist between the front wheels and the towbar when the locking pins are set. A towbar, which shall include the lunette eye, shall be hinged to permit vertical motion. The towbar shall be joined in two sections to permit lateral folding of the upper section. The vertical height in the stow position, with the upper section folded, shall not exceed the height of the Chassis (3.4.1). A mechanical lock shall be provided to retain the towbar in the vertical position.

3.4.3 Cable Harness

3.4.3.1 Function - The Cable Harness shall provide interconnection for Electronic Units when installed and shall connect external prime power to these units.

3.4.3.2 Form Factor - The Cable Harness shall consist of trunk and branch cables containing interconnecting wiring and coaxial cabling and terminating in connectors for attachment to the Electronic Units.

3.4.3.3 Connections - Connections to the Electrical Units shall be provided as listed in Table 1.



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TABLE 1

| Reference Designation | Receptacle Type | Function |
|---|---|---|
| 53A1P1 | 577R590H36 | Meter Panel Input |
| 53A1P2 | 577R569H01 | Meter Panel Deviation Signal Input |
| 53A1P3 | 577R569H01 | Meter Panel Deviation AFC Output |
| 53A1P4 | 577R569H01 | Meter Panel Deviation Output |
| 53A1P5 | 577R569H01 | Meter Panel 30 mc Calibration Output |
| 53A1P6 | 577R569H01 | Meter Panel Coding Signal Output |
| 53A3P2 | 577R569H01 | SA Analyzer Unit Signal Input |
| 53A4P2 | 575R516H03 | SA Filter Unit Noise Voltage Output |
| 53A5P1 | 577R590H23 | Electronic Drawer Power Input |
| 53A5P2 | 579R831H01 | Electronic Drawer 30 mc Output |
| 53A5P3 | 579R831H01 | Electronic Drawer Phase Detector Current Measure |
| 53A5P4 | 579R831H01 | Electronic Drawer Deviation Signal Measure |
| 53A5P5 | 579R831H01 | Electronic Drawer 30 mc Input |
| 53A5P6 | 579R831H01 | Electronic Drawer, AM Detector Current Measure |
| 53A5P7 | 579R831H01 | Electronic Drawer Signal Output |
| 53A5P8 | 579R831H01 | Electronic Drawer FM Signal Input |
| 53A5P9 | 579R831H01 | Electronic Drawer AM Signal Input |
| 53A5P10 | 579R831H01 | Electronic Drawer Frequency Discriminator Output |
| 53A5P11 | 579R831H01 | Electronic Drawer 30 cps Output |
| 53A5P12 | 577R590H41 | Electronic Drawer Signal/Control |
| 53A5P18 | 579R831H01 | Electronic Drawer Offset Oscillator AFC Input |
| 53A5P19 | 579R831H01 | Electronic Drawer Deviation Loop Input |
| 53A5P20 | 579R831H01 | Electronic Drawer Coding Audio Input |
| 53A6P1 | 577R569H01 | RF Panel AM Preamp Output |
| 53A6P2 | 577R569H01 | RF Panel AM Detector Current Output |
| 53A6P3 | 577R569H01 | RF Panel Deviation Output |
| 53A6P5 | 577R569H01 | RF Panel 30 mc Input |
| 53A6P7 | 577R590H36 | RF Panel Signal Ground Input |
| 53A6P8 | 577R590H36 | RF Panel Control Signal Input |
| 53A6P9 | 577R569H01 | RF Panel Phase Detector Current Output |
| 53A6P10 | 577R569H01 | RF Panel FM Output |
| 53A6P11 | 577R569H01 | RF Panel 30 cps Input |
| 53A7P1 | 577R590H45 | Power Supply (PS) Power and Signal Output |
| 53A7P3 | 577R569H01 | PS Deviation Input |
| 53A7P4 | 577R569H01 | PS Deviation Signal Output |
| 53A7P5 | 577R569H01 | PS Deviation Loop AFC Output |
| 53A11P1 | 577R831H01 | Meter Panel Power Meter Power Input |
| 53A11P2 | 577R831H01 | SA Analyzer Power Input |
| 53A11P3 | 577R831H01 | SA Converter Unit Power Input |



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3.4.4 Cable Assemblies

3.4.4.1 Power Cable - A Power Cable shall be furnished to convey single-phase, 50 to 440 cps electrical power to the Cart. The Cable shall be 240 ±6 inches in length and wired with No. 16 wire, per MIL-W-16878/4. A connector, MS 3126F12-3S, shall terminate one end and a connector, MS 24663, shall terminate the other end.

3.4.4.2 Cable Assembly, STALO (Stable Local Oscillator) - A Cable Assembly shall be provided to permit measurements to be made of the STALO signal. It shall connect the 30 megacycles (mc) loop output of the Electronic Units to the MCS and shall be coaxial cable RG-58/U, per MIL-C-17/28. Length shall not exceed 84 ±1 inches. An MS 35168-88E connector shall be provided on one end and a second MS 35168-88E connector with a BNC to TSM connector adapter shall be provided on the other end.

3.4.4.3 Cable Assembly, CW Illuminator - A Cable Assembly shall be provided to permit measurements to be made of either the CW Illuminator signal or the MCS STALO ranging signal. It shall connect either the CW Illuminator output or the MCS STALO and ranging signal output to the Electronic Units and shall be a flexible, metal jacket, coaxial cable. Length shall not exceed 84 ±1 inches. It shall have a 580R473H01 connector on each end with an adapter for type N connector to waveguide on the output end.

3.4.4.4 Cable, 30 MC/CW Illuminator - A signal jumper Cable shall be provided to couple a 30 mc output signal from one front panel connector to an adjacent front panel connector on the Electronic Drawer for CW Illuminator measurements. The Cable shall be RG58/U coaxial cable per MIL-C-17/28 and shall be 6±1/2 inches in length. The Cable shall terminate in an MS 35168-88E connector on each end.

3.4.4.5 Cable, 30 MC Calibration - A Cable shall be provided to connect the 30 mc output signal to the Spectrum Analyzer for calibration purposes. The Cable shall be RG58/U coaxial cable per MIL-C-17/28 and shall be 24 ±1 inches in length. The Cable shall terminate in an MS 35168-88E connector on each end.

3.4.4.6 Power Cable, Pseudo Sideband Modulator - A Power Cable shall be provided for use with the Pseudo Sideband Modulator for interconnection with the Power Supply Drawer. The cable shall be wired with no. 20 shielded and unshielded wire per MIL-W-16878/4 and shall be 57 ±1 inches in length. The connectors shall be 577R590H03 and 577R590H05.



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3.4.4.7 Signal Cable, Pseudo Sideband Modulator - A Signal Cable shall be provided to connect the output from the Pseudo Sideband Modulator to the Spectrum Analyzer input. The cable shall be RG58/U coaxial cable per MIL-C-17/28 and shall be 35 ±1 inches in length. The cable shall terminate in an MS 35168-88E connector on each end.

3.4.5 Waveguide Assemblies

3.4.5.1 Waveguide Assembly - A Waveguide Assembly shall be provided to connect input to output waveguide covers on the RF Panel of the Electronic Units. The assembly shall be an X-band waveguide of U-shape configuration terminating at both ends in a flange choke, per MIL-F-3922/27. Two clamps shall secure the assembly in place when attached to the RF Panel. Nominal size shall be 2 inches from choke centerline to choke centerline and 2-1/2 inches from choke face to clamp top.

3.4.5.2 Waveguide to Coaxial Adapter - A Waveguide to Coaxial Adapter shall be furnished to provide a transition from the CW Illuminator Cable Assembly (3.4.4.3) to the input flanges of the RF Panel. The Adapter shall consist of a section of RG67/U waveguide, per MIL-W-85C of 5-3/4 inches nominal length terminated on one end by a UG136B/U modified flange per MIL-F-3922 and on the other end by a type N connector per MIL-C-71.

3.4.5.3 Waveguide Extension - A Waveguide Extension shall be provided to connect the Pseudo Sideband Modulator to the RF Panel. The Waveguide Extension shall be an X-band, straight waveguide section of nominal six inches in length. Each end shall be terminated in a flange choke, per MIL-F-3922/27.

4. QUALITY ASSURANCE PROVISIONS

4.1 General - Equipments shall be subjected to the following tests to determine compliance with all the applicable requirements.

(a) Preproduction Tests

(b) Acceptance Tests

(c) Life Tests

NOTE: Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the



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inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction Tests - Preproduction tests shall be made on one or more equipments representative of the production equipments to be supplied under the contract. Preproduction tests shall consist of the following:

(a) Contractor's Demonstration Tests

(b) Service Approval Tests

4.2.1 Contractor's Demonstration Tests - Contractor's demonstration tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of (4.5). The Government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the service of a commerical testing laboratory satisfactory to the procuring activity.

4.2.2 Service Approval Tests - At the completion of the contractor's demonstration test and when requested by the procuring activity the equipment shall be delivered to the Government for additional testing. This additional testing may consist of duplicating tests previously conducted and such other tests as are deemed necessary to determine compliance with all applicable design and performance requirements.

4.2.3 Scope of Tests - Preproduction tests shall include all tests necessary to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests shall include environmental testing in accordance with procedures of MIL-T-5422 with test limits modified to correspond with the limits specificed herein and interference tests in accordance with MIL-I-6181.

4.2.4 Preproduction Approval - Approval of the preproduction models shall be by the Naval Air Systems Command upon satisfactory completion of all tests. All preproduction models, except those consumed in testing or those rendered irrepairable for their intended use, shall be returned to the Contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction models shall not be considered as equipments under this contract; however, such models may be reworked and submitted for acceptance as production equipments. Incorporation of design changes, in delivered production equipments, found to be necessary as a



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result of the preproduction test program, shall be the subject of separate and independent negotiations on an individual basis.

4.3 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the Government Inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(a) Individual Tests

(b) Sampling Tests

(c) Special Tests

4.3.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(a) Examination of Product

(b) Operational Test

(c) Reliability Test

4.3.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material, workmanship and interchangeability requirements have been met.

4.3.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.3.1.3 Reliability Test - Each equipment shall be operated for a period of 6 hours without failure, under the conditions specified herein. Only those adjustments will be permitted which can be made by using such controls and adjustments as are accessible to the operator during the normal use of the equipment.



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| | |
|---|---|
| Temperature | Room Ambient |
| Humidity | Room Ambient |
| Vibration (10 minutes) | Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and a minimum double amplitude of 0.036 inch. |

The equipment shall be vibrated (without vibration isolators between the equipment and test fixtures) for a period of 10 minutes prior to the beginning of the 6-hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to that plane for 5 minutes. Where it is not feasible to vibrate the equipment in 2 directions the vertical direction shall be used. During the 6-hour period of operation following the 10-minute vibration period, the equipment shall be cycled periodically through its phases of operation. Should a failure occur, it should be repaired and tests started over, except the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The 6-hour period specified above may be composed of two 3-hour periods to conform with standard working hours.

4.3.2 Sampling Tests - Sampling tests shall be conducted on complete equipments which have successfully passed the individual tests. The following equipments shall be selected for Sampling tests:

| Quantity of Equipments Offered for Acceptance | Quantity to be Tested |
|---|---|
| First 10 | 1 |
| 11 to 100 | 2 additional |
| 101 to 200 | 1 additional |
| Over 200 | 1 for each additional 200 or fraction thereof |



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4.3.2.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests after successful completion of the tests of 4.3.1.

(a) Complete operational tests at ambient room conditions making all necessary measurements to assure that all applicable specification requirements have been met.

(b) Operational test at environmental conditions when the results from the preproduction test indicate that these environmental conditions cause degradation(s) which result in marginal or unsatisfactory performance.

4.3.3 Special Tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling tests.

4.3.3.1 Special Test Schedule - Selection of equipments for special tests shall be made as follows:

(a) On an early production equipment.

(b) On an equipment midway of the production run.

(c) On an early equipment after an extensive engineering change.

(d) When failure reports or other information indicate additional equipment should be tested. (This will be determined by the procuring activity.)

Exceptions from the above may be necessary when the number of equipments procured under contract or other factors affecting performance may require a variation to assure satisfactory control. The maximum number of special tests shall be defined by the procuring agency at the time of request for bids.

4.3.3.2 Scope of Test - Special tests shall consist of such tests as are selected by the procuring activity. Test procedures previously approved for the preproduction and sampling tests shall be used where applicable. As a minimum, the following tests shall be made, except that for those equipments selected as a result of design changes and failure reports, only such tests required to check the characteristics in question need be conducted:



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(a) Temperature - Altitude Tests

(b) Vibration Tests (90 minutes cycling in each of 3 planes)

(c) Humidity Tests

4.3.4 Equipment Failure - Should a failure occur during either the sampling or special tests, the following action shall be taken:

(a) Determine the cause of failure.

(b) Determine if the failure is an isolated case or design defect.

(c) Where practical, include a test in the individual test to check all equipment for this requirement until assurance is obtained that the defect has been satisfactorily corrected.

4.4 Life Test - A 300-hour life test shall be conducted at the contractor's plant on an early equipment representative of the current production. The life test shall be performed under the conditions specified in 4.4.1. The life tests sample shall be selected by the Government Inspector in accordance with the following: (Equipments which have successfully passed the sampling tests or special tests may be selected for life tests.)

| Quantity of Equipments Offered for Acceptance | Quantity to be Tested |
|---|---|
| First 25 | 1 |
| 26 to 200 | 1 additional |
| 201 to 500 | 1 additional |
| Over 500 | 1 for each additional 500 or fraction thereof |

4.4.1 Test Conditions - The life test shall be conducted under the following conditions:

| | |
|---|---|
| Temperature | Room Ambient |
| Altitude | Normal Ground |
| Humidity | Room Ambient |



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4.4.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off and put through its various phases of operation.

4.4.3 Performance Check - At approximately 8-hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

4.4.4 Reconditioning of Life Test Samples - An equipment which has been subject to the life test shall be reconditioned as follows:

(a) On completion of the life test, the equipment shall be reworked by the contractor by replacing all "wear" items. The "wear" items shall be determined by agreement between the contractor and the procuring activity.

(b) After reworking, the contractor shall resubmit the equipment for acceptance.

4.5 Test Procedure - Preproduction test, acceptance tests and life tests shall be performed in accordance with a test plan and procedures mutually agreeable to the procuring activity and the contractor. The test plan shall be prepared by the contractor. MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved procedures are available from previous contracts, the procuring activity will furnish the procedures to be used for testing.

4.6 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply to the best of his knowledge and belief, with all applicable requirements.

4.7 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and be resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the Government Inspector. Units rejected after retest shall not be resubmitted without the specific approval of the procuring activity.



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4.8 Equipment Changes - After approval of the preproduction model, no changes which affect weight, installation interchangeability or interchangeability of maintenance parts of assemblies shall be incorporated in the equipment unless approved by the procuring activity. To obtain procuring activity approval, the contractor shall submit an Engineering Change Proposal (ECP) giving the necessary information concerning the change. This shall apply for either a procuring activity recommended change or a contractor's proposed change. When required, the contractor shall also prepare a Material Change (MC) and a Material Bulletin (MB) to cover the approved change. Engineering Change Proposals, Material Changes and Material Bulletins shall be prepared in accordance with MIL-T-23336.

## 5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with MIL-E-17555.

## 6. NOTES

6.1 Ordering Data - Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(a) Title, Number, and date of this specification.

(b) Selection of applicable levels of packaging and packing (see 5.1).

6.2 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(a) Contract - The contract shall have precedence over any specifications.

(b) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(c) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.



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6.3 Notation of Revisions - A bar in the margin denotes a change made in that paragraph from the previous version of the document. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in those notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written irrespective of the marginal notations and relationship to the last previous issue.

6.4 Availability of Documents - When requesting specifications, standards, drawings, and publications refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code 105, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.



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# SPECIFICATION ANALYSIS SHEET

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INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-C-81532(AS) CART, TEST EQUIPMENT

| ORGANIZATION *(Of submitter)* | CITY AND STATE |
|---|---|
| | |

| CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT |
|---|---|---|
| | | $ |

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

| SUBMITTED BY *(Printed or typed name and activity)* | DATE |
|---|---|
| | |

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